Software Engineering

# System Design Description

# Momento

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Riga 2026

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# INTRODUCTION

## DOCUMENT PURPOSE

This document contains the design of the Momento mobile application and a description of its structure and operation. The document is intended for system developers, testers, and project evaluators.

The design document defines the modular structure of the system, the decomposition of processes and data, inter-module and inter-process dependencies, interface specifications, and detailed decision logic. It serves as the primary technical reference for implementation of the Momento system according to the requirements defined in the System Requirements Specification.

## SCOPE OF OPERATION

Momento is a location-based mobile application that enables users to drop digital time capsules - called momentos - at specific GPS coordinates, and to discover capsules dropped by other users by physically visiting those locations. The system targets two user roles that any registered user may occupy simultaneously: Creators and Discoverers.

The system is developed as a mobile-first application for iOS (14+) and Android (10+), using React Native with Expo on the frontend and Java with Spring Boot on the backend. Data persistence is handled by a PostgreSQL database with the PostGIS extension for geospatial queries. Media files are stored in AWS S3 or Firebase Storage. Authentication is provided by Firebase Authentication, and push notifications are delivered via Firebase Cloud Messaging (FCM).

## DEFINITIONS AND ABBREVIATIONS

*Table 1*

| Term | Explanation |
|---|---|
| **User** | A registered person who interacts with the Momento application as a Creator, Discoverer, or both. |
| **Creator** | A user who drops digital time capsules at GPS-specified locations. |
| **Discoverer** | A user who physically visits capsule locations to unlock and view their content. |
| **Momento / Capsule** | A digital time capsule containing media (photo, voice note, or text) anchored to a GPS coordinate. |
| **Design Document** | A document describing the system structure and operation in sufficient detail for implementation. |
| **Decomposition** | The division of a system into multiple components such as modules, processes, or data structures. |
| **PostgreSQL / PostGIS** | A relational database system with the PostGIS extension enabling geospatial queries. |
| **Module** | A component of the system that performs a specific function and operates as part of the larger system. |

| | |
|---|---|
| **Process** | An identifiable sequence of data-processing activities within the system with a defined initial state and possible end states. |
| **FCM** | Firebase Cloud Messaging - a cross-platform service for sending push notifications. |
| **GPS** | Global Positioning System - used to determine and record precise geographic coordinates. |
| **JWT** | JSON Web Token - a compact, URL-safe token used for authentication session management. |
| **API** | Application Programming Interface - a set of rules for communication between system components. |
| **S3** | Amazon Simple Storage Service - cloud object storage used for media files. |

RELATIONSHIP WITH OTHER DOCUMENTS: This document is based on the Momento System Requirements Specification (SRS) and the Momento Security Vulnerability Analysis, both dated Riga 2026.

# DESCRIPTION OF DECOMPOSITION

## MODULAR DECOMPOSITION

The Momento system is decomposed into four primary modules, each responsible for a distinct area of functionality. The Capsule Processing Module is further divided into three submodules handling capsule creation, discovery, and expiry logic respectively.

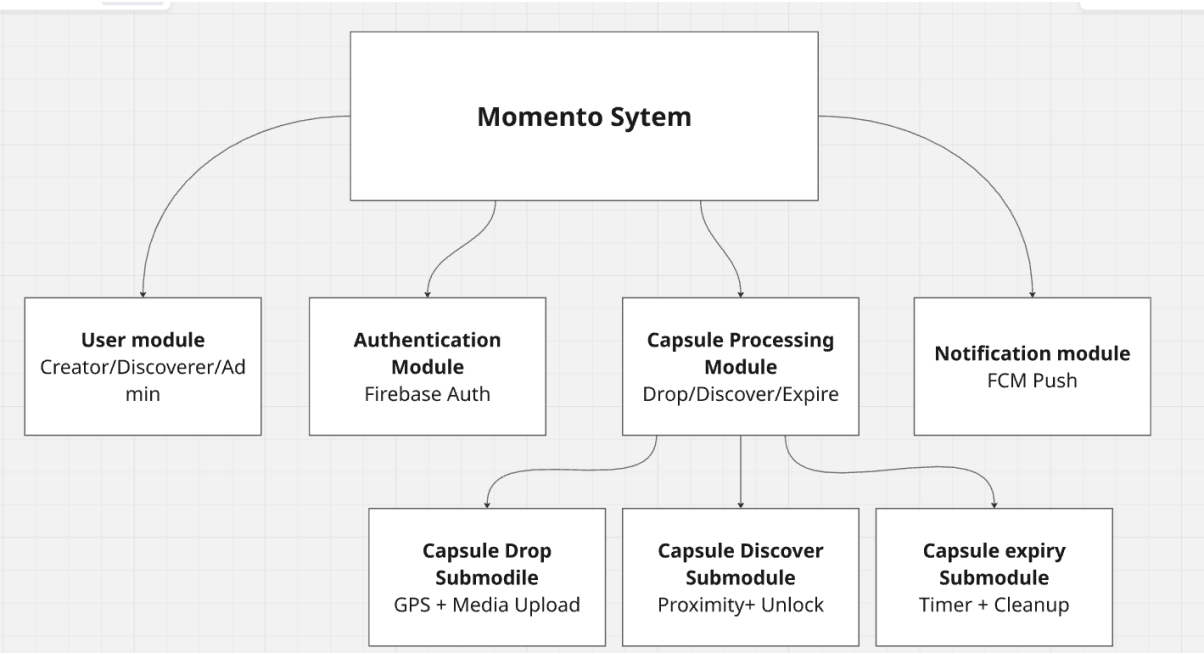


*Image 1 - Modular Decomposition*

## MODULE DECOMPOSITION LIST

*Table 2*

| | |
|---|---|
| **Name** | User Module |
| **Type** | User Module |
| **Purpose** | Data input, service access, and role-based interaction. |
| **Function** | Allows registered users to act as Creators or Discoverers. Provides access to capsule creation, map view, discovery history, user profile, and the points system. |

*Table 3*

| | |
|---|---|
| **Name** | Authentication Module |
| **Type** | Authentication Module |
| **Purpose** | User identity verification and session management. |
| **Function** | Handles user registration and login via email/password or third-party providers (Google, Apple) through Firebase Authentication. Issues and validates JWT session tokens. |

*Table 4*

| | |
|---|---|
| **Name** | Capsule Processing Module |
| **Type** | Capsule Processing Module |
| **Purpose** | Central coordination of all capsule-related operations. |
| **Function** | Selects the appropriate submodule depending on the required operation: dropping a capsule, discovering a capsule, or processing capsule expiry. |

*Table 5*

| | |
|---|---|
| **Name** | Capsule Drop Submodule |
| **Type** | Capsule Processing Module - Submodule |
| **Purpose** | Creation and storage of new capsules. |
| **Function** | Validates user input, captures GPS coordinates, stores media content in cloud storage, saves the capsule record (including location, media URL, creator ID, and expiry timestamp) to the database, and awards points to the creator. |

*Table 6*

| | |
|---|---|
| **Name** | Capsule Discovery Submodule |
| **Type** | Capsule Processing Module - Submodule |
| **Purpose** | Detection and unlocking of capsules based on user proximity. |
| **Function** | Continuously monitors the user GPS position, queries the database for nearby undiscovered capsules, unlocks capsule content when the user enters the proximity radius, records the discovery event, extends the capsule expiry by 30 days, awards points to the discoverer, and sends a push notification to the capsule creator. |

*Table 7*

| | |
|---|---|
| **Name** | Capsule Expiry Submodule |
| **Type** | Capsule Processing Module - Submodule |
| **Purpose** | Automated lifecycle management of capsule expiry. |
| **Function** | Runs a scheduled timer check against all active capsules. Permanently deletes capsules whose expiry timestamp has elapsed without a discovery, and removes their associated media from cloud storage. Sends expiry warning notifications 3 days before expiry. |

*Table 8*

| | |
|---|---|
| **Name** | Notification Module |

| | |
|---|---|
| **Type** | Notification Module |
| **Purpose** | Delivery of push notifications to users. |
| **Function** | Communicates with Firebase Cloud Messaging (FCM) to deliver push notifications for discovery events and capsule expiry warnings. |

## PROCESS DECOMPOSITION

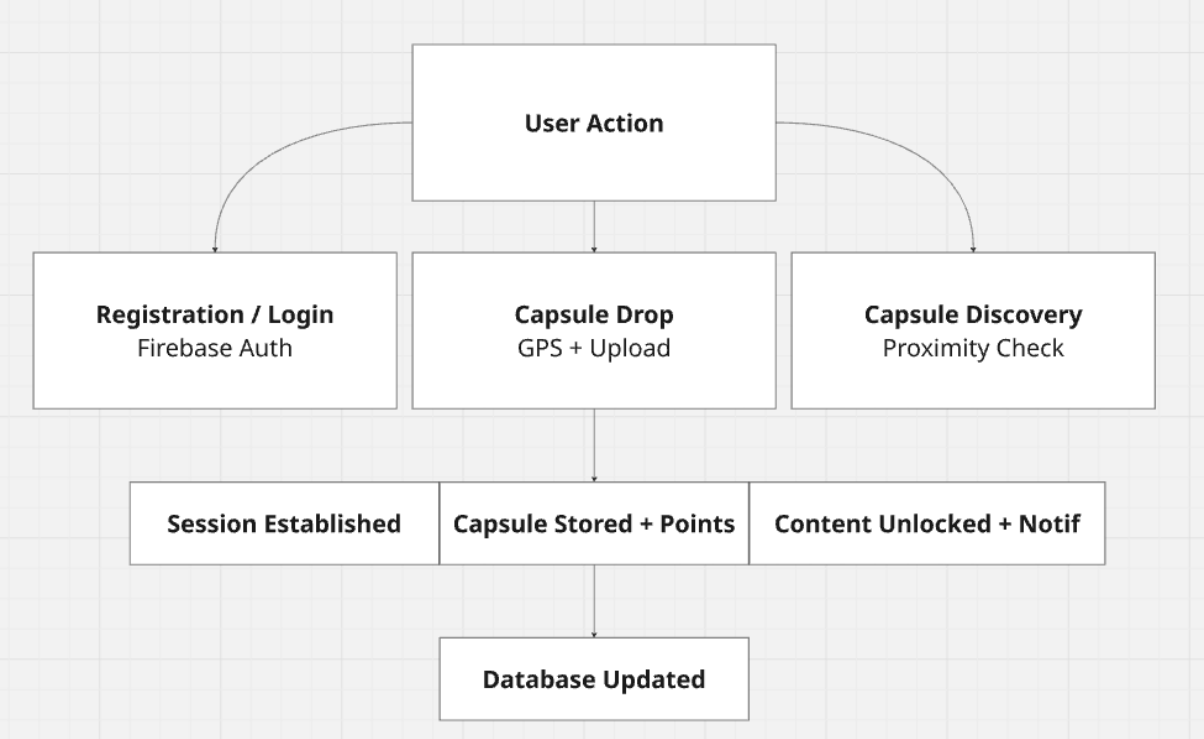


*Image 2 - Process Decomposition*

*Table 9*

| Module | Process Name | Process Description |
|---|---|---|
| User Module | **Account Registration** | The user provides email and password or selects a social provider. The system checks for existing accounts, creates a new profile, assigns the default role, and establishes a session. |
| User Module | **User Login** | The user enters credentials. Firebase Authentication validates them and issues a JWT session token on success. An error is shown on failure. |
| User Module | **Map View** | Retrieves the user GPS position and queries the database for nearby undiscovered capsule markers, displaying them on the map without revealing content. |

| | | |
|---|---|---|
| Capsule Drop Submodule | **Capsule Creation** | The user attaches content (photo, voice note, or text), the app captures GPS coordinates, the backend validates input, stores media in cloud storage, saves the capsule record, creates a map marker, and awards points. |
| Capsule Discovery Submodule | **Proximity Check** | Continuously compares the user GPS position against capsule locations stored in PostGIS. Triggers unlock when the user enters the capsule proximity radius. |
| Capsule Discovery Submodule | **Capsule Unlock** | Unlocks and displays capsule content, extends expiry by 30 days, records the discovery event, awards points to the discoverer, and sends FCM notification to the creator. |
| Capsule Expiry Submodule | **Expiry Check** | A scheduled background process that checks all active capsules. Deletes expired capsules and their media, and sends 3-day warning notifications. |
| Notification Module | **Push Notification Delivery** | Communicates with FCM to deliver opaque push notifications. FCM tokens are managed server-side and refreshed on logout. |

## DATA DECOMPOSITION

| | |
|---|---|
| **users**<br>- user_id (PK)<br>- email<br>- username<br>- points_total<br>- created_at<br>- auth_provider | **capsules**<br>- capsule_id (PK)<br>- creator_id (FK -> users)<br>- location (PostGIS POINT)<br>- media_url<br>- content_type<br>- expiry_at<br>- created_at |
| **discoveries**<br>- discovery_id (PK)<br>- capsule_id (FK -> capsules)<br>- discoverer_id (FK -> users)<br>- discovered_at<br>- points_awarded | **media**<br>- media_id (PK)<br>- capsule_id (FK -> capsules)<br>- storage_url<br>- media_type (photo/voice/text)<br>- size_bytes<br>- uploaded_at |

*Image 3 - Data Decomposition (Entity Overview)*

*Table 10*

| Table Name | Description |
|---|---|
| **users** | Stores registered user accounts including user ID, email, username, authentication provider, total points, and account creation timestamp. |
| **capsules** | Stores all dropped capsules including capsule ID, creator ID (foreign key to users), GPS location as a PostGIS POINT geometry, media storage URL, content type, expiry timestamp, and creation timestamp. |
| **discoveries** | Records each discovery event with discovery ID, capsule ID (foreign key to capsules), discoverer ID (foreign key to users), discovery timestamp, and points awarded. |
| **media** | Tracks uploaded media files with media ID, capsule ID (foreign key to capsules), storage URL, media type (photo, voice, or text), file size in bytes, and upload timestamp. |

# DESCRIPTION OF DEPENDENCES

## INTER-MODULE DEPENDENCIES

The diagram below shows the data and control flow between the four primary modules of the Momento system. The User Module is the entry point for all interactions. The Authentication Module is invoked first to establish a session before any other module is accessible. The Capsule Processing Module communicates with the database for all read and write operations. The Notification Module is called by the Capsule Processing Module to dispatch push notifications upon discovery or expiry events.

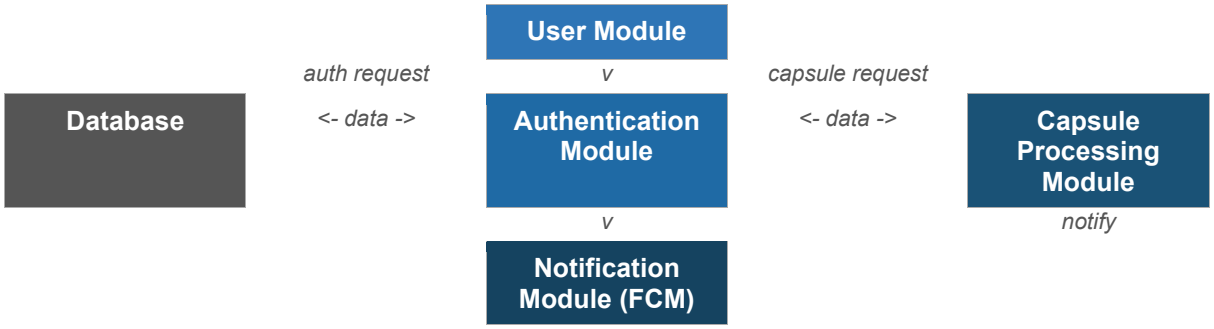


*Image 4 - Inter-module Dependencies*

## INTER-PROCESS DEPENDENCIES

The processes within the Momento system follow a sequential dependency model. User registration and login must complete before any capsule operation can be performed. Capsule creation depends on GPS coordinates being available and media upload completing successfully before the database record is saved. Capsule discovery depends on the proximity check process completing and confirming that the capsule belongs to a different user. The expiry process runs independently on a scheduled basis and does not depend on any user-initiated process.

*Table 11*

| Process | Depends On | Dependency Description |
|---|---|---|
| **User Login** | Account Registration | A user can only log in if a registered account exists. Registration must precede the first login attempt. |
| **Map View** | User Login | The map view is only accessible after a valid session token has been issued by the authentication process. |
| **Capsule Creation** | Map View / GPS | A capsule can only be dropped if the user is authenticated and the app has captured valid GPS coordinates. |
| **Media Upload** | Capsule Creation Input | Media upload to cloud storage is triggered only after the user confirms capsule submission and input validation passes. |
| **Database Record Save** | Media Upload | The capsule record (including media URL) is saved to the database only |

| | | |
|---|---|---|
| | | after media upload completes successfully. |
| **Proximity Check** | User Login + GPS | Proximity checking requires both an active user session and continuous GPS position updates. |
| **Capsule Unlock** | Proximity Check | Content is unlocked only after the proximity check confirms the user is within the capsule radius and the capsule belongs to a different user. |
| **Discovery Record Save** | Capsule Unlock | The discovery event is recorded in the database only after content has been successfully unlocked. |
| **Push Notification** | Discovery Record Save | The FCM notification to the capsule creator is dispatched only after the discovery record is committed to the database. |
| **Expiry Deletion** | Scheduled Timer Check | Capsule deletion runs as an independent scheduled job and does not depend on any user-initiated process. |

## DATA DEPENDENCY DESCRIPTION

The data model of the Momento system is structured around four core tables. The users table is the root entity - all other tables reference it via foreign keys. The capsules table stores geospatial capsule data and is linked to the users table through the creator_id foreign key. The media table stores individual media files and is linked to the capsules table through the capsule_id foreign key. The discoveries table records discovery events and references both the capsules and users tables.

*Table 12*

| Source Table | Target Table | Foreign Key / Relationship |
|---|---|---|
| **capsules** | users | creator_id - links each capsule to the user who created it. |
| **discoveries** | capsules | capsule_id - links each discovery event to the discovered capsule. |
| **discoveries** | users | discoverer_id - links each discovery event to the user who discovered it. |
| **media** | capsules | capsule_id - links each media file to the capsule it belongs to. |

# INTERFACE DESCRIPTION

## MODULE INTERFACES

*Table 13*

| Module Name | User Module |
|---|---|
| **Input Data** | - User credentials (email, password, or OAuth token)<br>- GPS coordinates<br>- Capsule media content (photo, voice note, or text)<br>- Capsule ID (for deletion or viewing) |
| **Output Data** | - Authenticated session token<br>- Map view with capsule markers<br>- Capsule content (on discovery)<br>- User profile data and points balance<br>- Error or confirmation messages |

*Table 14*

| Module Name | Authentication Module |
|---|---|
| **Input Data** | - Email and password<br>- OAuth provider token (Google, Apple)<br>- JWT token (for validation on each request) |
| **Output Data** | - JWT session token (on successful login or registration)<br>- Validation result (valid or invalid token)<br>- Error message (on authentication failure) |

*Table 15*

| Module Name | Capsule Processing Module |
|---|---|
| **Input Data** | - Capsule media content and GPS coordinates (for drop)<br>- User GPS position (for proximity check)<br>- Capsule ID (for deletion or status query)<br>- Scheduled timer signal (for expiry check) |
| **Output Data** | - Capsule creation confirmation and map marker<br>- Unlocked capsule content (on discovery)<br>- List of user capsules with status<br>- List of discovery history<br>- Expiry deletion confirmation<br>- FCM notification trigger |

*Table 16*

| Module Name | Notification Module |
|---|---|
| **Input Data** | - FCM token (target device) |

| | |
|---|---|
| | - Notification type (discovery event or expiry warning)<br>- Capsule metadata (opaque reference, no content) |
| **Output Data** | - Push notification delivered to target device<br>- Delivery status (success or failure) |

## PROCESS INTERFACES

*Table 17*

| Process Name | User Registration |
|---|---|
| **Input Data** | Email address, password (or OAuth token from Google or Apple), and username. |
| **Output Data** | Confirmation of account creation and a JWT session token. An error message is returned if the email already exists. |

*Table 18*

| Process Name | User Login |
|---|---|
| **Input Data** | Email and password (or OAuth token). |
| **Output Data** | JWT session token on success. An error message on failure. |

*Table 19*

| Process Name | Capsule Drop |
|---|---|
| **Input Data** | Media content (photo, voice note, or text), current GPS coordinates captured by the app, and optional text message. |
| **Output Data** | Confirmation of capsule creation, a map marker at the capsule location, and points awarded to the creator. |

*Table 20*

| Process Name | Proximity Check |
|---|---|
| **Input Data** | User current GPS coordinates (continuously updated). |
| **Output Data** | List of capsule IDs within the user proximity radius, or an empty result if no nearby undiscovered capsules are found. |

*Table 21*

| Process Name | Capsule Unlock |
|---|---|
| **Input Data** | Capsule ID (identified by proximity check), authenticated user ID. |
| **Output Data** | Unlocked capsule content (media and text), expiry extension confirmation, discovery event record, and FCM notification dispatch. |

*Table 22*

| Process Name | Expiry Check |
|---|---|
| **Input Data** | Scheduled timer signal (no user input). |
| **Output Data** | List of deleted capsule IDs and confirmation of media removal from cloud storage. FCM expiry warning notifications dispatched for capsules expiring within 3 days. |

*Table 23*

| Process Name | Push Notification Dispatch |
|---|---|
| **Input Data** | FCM device token, notification type, and opaque capsule reference. |
| **Output Data** | Notification delivery confirmation from FCM (success or failure status). |

# DETAILED DESIGN DESCRIPTION

## DECISION TABLES

### Authentication Check

*Table 24*

| Conditions / Actions | 1 | 2 |
|---|---|---|
| **C1 - Are credentials valid?** | 0 | 1 |
| **Actions** | | |
| **D1 - Show error message and prompt retry** | X | |
| **D2 - Issue JWT token and redirect to Map View** | | X |

### Capsule Creation Input Validation

*Table 25*

| Conditions / Actions | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **C1 - Media content attached?** | 0 | 0 | 1 | 1 |
| **C2 - GPS coordinates captured?** | 0 | 1 | 0 | 1 |
| **Actions** | | | | |
| **D1 - Show error: No content attached** | X | X | | |
| **D2 - Show error: GPS unavailable** | | | X | |
| **D3 - Upload media and save capsule** | | | | X |

### Capsule Discovery - Proximity and Ownership Check

*Table 26*

| Conditions / Actions | 1 | 2 |
|---|---|---|
| **C1 - User within proximity radius?** | 0 | 1 |
| **C2 - Capsule owned by user?** | - | 0 / 1 |
| **Actions** | | |
| **D1 - Continue monitoring (no action)** | X | |
| **D2 - Show message: Cannot discover own capsule** | | X (own) |
| **D3 - Unlock capsule and trigger discovery flow** | | X (other) |

### Capsule Expiry Check

*Table 27*

| Conditions / Actions | 1 | 2 | 3 |
|---|---|---|---|
| **C1 - Capsule expiry timestamp elapsed?** | 0 | 0 | 1 |
| **C2 - Expiry within 3 days?** | 0 | 1 | - |
| **Actions** | | | |
| **D1 - No action (capsule active)** | X | | |
| **D2 - Send expiry warning notification via FCM** | | X | |
| **D3 - Delete capsule and remove media from storage** | | | X |

### Duplicate Capsule Coordinate Rule (24-hour restriction)

*Table 28*

| Conditions / Actions | 1 | 2 |
|---|---|---|
| **C1 - User has existing capsule at same coordinates within 24h?** | 1 | 0 |
| **Actions** | | |
| **D1 - Reject request and show error: Duplicate location** | X | |
| **D2 - Proceed with capsule creation** | | X |

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